

## 4.1 Forces and equilibrium

### Mechanics · 2026 examination · Concept guide and original exam-style practice

#### Understand the concept

Draw a free-body diagram and resolve each force into horizontal and vertical components. Equilibrium requires zero resultant in both directions. Friction is bounded by  $\mu R$  for limiting friction; otherwise its actual magnitude is determined by equilibrium/dynamics, not automatically  $\mu R$ .

#### Worked example: follow the reasoning

A 10 N force at  $30^\circ$  above horizontal has horizontal component  $5\sqrt{3}$  N and vertical component 5 N.

#### Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A block weighs 20 N on horizontal surface and is pulled horizontally by 6 N while at rest. State friction and reaction, assuming no other vertical forces. [2]

Final answer: \_\_\_\_\_

## Answers and explanation

### Question 1 [2 marks]

Friction = 6 N opposite pull [1]; normal reaction = 20 N [1].

### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.